PLANCKS 2026

QUESTION BOOKLET



Rules and regulations

Dear participants,
Here are some of the most important rules of the contest.

- The langugage used in the competition is English.
- The contest consists of 9 problems. All problems are worth 10 points, except for the final question which is worth only 5 points. Subdivisions of points are indicated in the problems.
- Each problem must be handed separately. No sheet of paper should contain solutions to two seperate problems. This also means that you are not allowed to use the front of a sheet for one problem and the back for another problem. Please mark the header of each sheet with your team name and write the problem number in the top right corner. It is necessary to mark each sheet so that it is scored.
- When a problem is unclear, a participant may request clarification through the invigilator. If the response is relevant to all teams, the jury will provide this information to the other teams.
- Participants are allowed to use a dictionary: English to your native language.
- Participants are allowed to use a non-programmable, not-graph calculator (scientific is okay).
- No books or other sources, except for this Question Booklet and a dictionary, are to be consulted during the competition.
- The use of hardware (including phones, tablets etc.) is not approved, except for medical equipment. Please leave your phones in an envelope.
- Participants may not leave the exam room until one hour after the start of the competition. Afterwards, they must obtain permission from the invigilator to use the restroom.
- The jury has the right to disqualify teams for misbehavior or breaking the rules.
- In situations to which no rule applies, the Organizing Committee decides.

Good luck and enjoy!

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1 Cheerios effect – Breakfast physics [10 points]

Prof. Dr. Hanneke Gelderblom

If you carefully observe your breakfast cereals in the morning, you may have notices that floating cereals often clump together, or stick to the wall of your bowl, see figure 1. This physical phenomenon goes by the name 'Cheerios effect' [1]. In this problem we will investigate this attraction by modelling the Cheerios as spheres with density ρ_c and radius R, floating in milk with a density $\rho_m > \rho_c$ and surface tension γ; see figure 1. The angle of contact between the Cheerios and the milk is a fixed angle $\theta < 90°$ depending on the properties of the Cheerios and the milk.

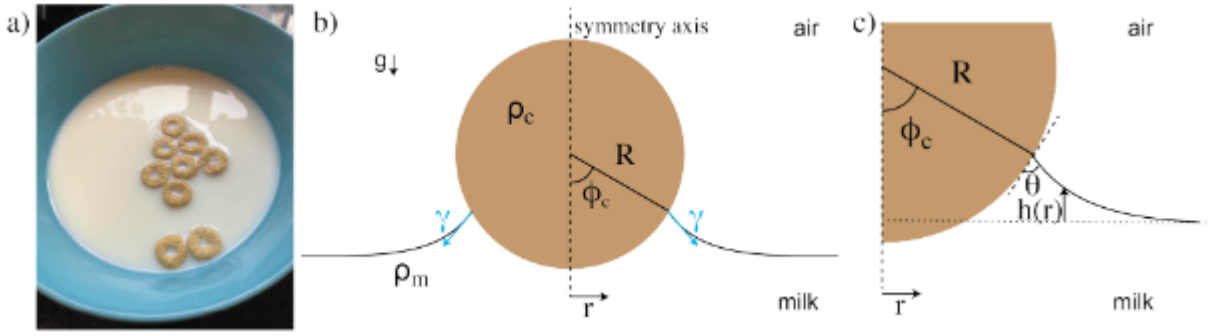


Figure 1: (a) Cheerios in your breakfast bowl clump together. (b) Our simplified Cheerio with density ρ_c and radius R floating in milk of density ρ_m. The gravitational acceleration g and the surface tension force per unit length γ that is pulling parallel to the interface are shown. The cylindrical coordinate r is indicated, with $r_c = R\sin(\phi_c)$ the location of the milk-Cheerio-air contact line. (c) Zoom of the contact-line region showing contact angle θ of the Cheerios with the liquid-air interface, and the elevated height $h(r)$ of the milk-air interface close to the Cheerio with respect to the horizontal.

1. *[3 points]:* Show that a vertical force balance on a floating spherical Cheerio yields

$$\sin(\phi_c)\frac{h_c'}{\sqrt{1+{h_c'}^2}} = B\left(\frac{2D-1}{3} + \frac{1}{2}\cos(\phi_c) - \frac{1}{6}\cos^3(\phi_c) + \frac{h_c}{2R}\sin^2(\phi_c)\right)$$

   with $h_c' = \frac{dh}{dr}|_{r=r_c}$ the slope of the milk-air interface at the three-phase contact-line between the Cheerio, the milk and surrounding air, which is located at $r_c = R\sin(\phi_c)$. Dimensionless parameter $B = \frac{\rho_m g R^2}{\gamma}$ is the Bond number and $D = \frac{\rho_c}{\rho_m}$ a density ratio. Hints:

   i. Surface tension γ (units $[\mathrm{N}\cdot\mathrm{m}^{-1}]$) is a force per unit length of the three-phase contact line, acting parallel to the milk-air interface (see figure 1).

   ii. One can show[2] that the volume of liquid displaced by a partially submerged body is equal to the volume bounded by the wetted surface of that body, a vertical cylinder, and the original horizontal air-liquid interface, and is given by

$$V_{\mathrm{dis}} = \pi R^3\left(-\frac{h_c}{R}\sin^2(\phi_c) + \frac{2}{3} - \cos(\phi_c) + \frac{1}{3}\cos^3(\phi_c)\right)$$

   with $h_c = h(r_c)$.

2. *[2 points]:* Show that for small interface deformations (i.e. $h_c \ll R$, $h_c' \ll 1$) this expression may be simplified to

$$\sin(\phi_c)h_c' = B\Sigma$$

[1] Vella and Mahadevan, *The "Cheerios effect"*, Am.J. Phys. 73 (9), 2005
[2] Keller, *Surface tension force on a partly submerged body*, Phys. Fluids 10(11), 1998

with

$$\Sigma = \frac{2D-1}{3} - \frac{1}{2}\cos(\theta) + \frac{1}{6}\cos^3(\theta)$$

the dimensionless resultant weight of the Cheerios.

Now consider two floating Cheerios approaching each other. Each Cheerio causes an interfacial deformation $h(r)$ of the milk-air interface in its proximity. This interfacial deformation is determined by the balance between the local capillary and hydrostatic pressures in the milk $\gamma\nabla^2 h = \rho_m g h$. One can show that as a result, the interface shape close to an isolated Cheerio is approximately given by

$$h(r) = -h'_c \sin(\phi_c) R K_0\left(\frac{r}{L}\right)$$

where $L = \sqrt{\gamma/\rho_m g}$ is the capillary length scale, and $K_n(x)$ is the nth-order modified Bessel function of the second kind, which is real and positive-valued for $n \geq 0, x > 0$ and tends to zero as $|x| \to \infty$. Furthermore $\frac{dK_0}{dx} = -K_1(x)$. In the following, you may assume that (to leading order) the presence of Cheerio 1 does not affect the shape of the interface close to Cheerio 2, and vice versa.

3. *[3 points]:* Show that the change in gravitational potential energy of Cheerio 1 by interacting with the deformed interface caused by Cheerio 2 is given by

$$E(\ell) = -2\pi\gamma R^2 B^2 \Sigma^2 K_0\left(\frac{\ell}{L}\right)$$

with ℓ the horizontal distance between the centers of the two Cheerios.

4. *[1 point]:* Determine the horizontal force of interaction $F(\ell)$ between the two Cheerios and explain why they will attract each other.

Now suppose that next to a Cheerio, you also carefully float a pushpin (with a density $\rho_p > \rho_m$, so its weight will be carried by the surface tension force) at the surface of your milk. Assume the pushpin has the same wetting properties as the Cheerio (i.e. same contact angle with milk).

5. *[1 point]:* If you bring this pushpin close to a Cheerio, will they also attract each other and clump together? Why (not)?

End of question.

2 Partial coherence and 3D imaging [10 marks]

Prof. Dr. Andrea Fiore

Optical coherence tomography (OCT) is a medical imaging technique based on the partial coherence of light. It allows resolving layers at different depths below the surface of a tissue, and it is for example used to make a cross-sectional image of the retina.

Consider the simplified OCT system shown in figure 2.

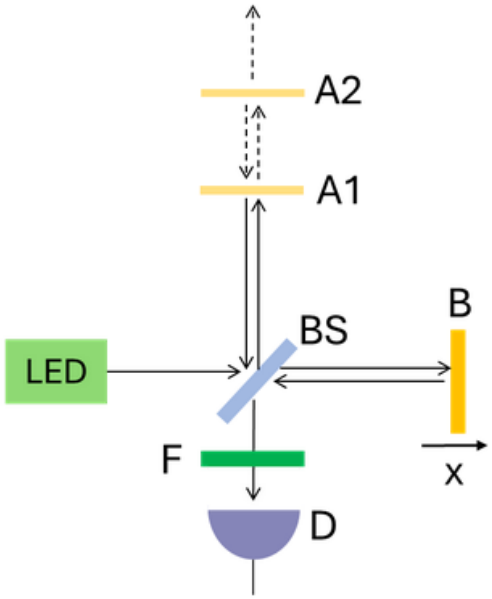


Figure 2: A simplified OCT system.

The source is a light-emitting diode (LED) which emits chaotic light with a power spectral density given by a single Lorentzian line centred at a wavelength $\lambda_0 = 1300$ nm, with a full-width half-maximum of 10 nm. **BS** is a 50/50 beamsplitter with equal transmittance and reflectance, **B** is a perfect mirror and **A1**, **A2** are two semireflecting interfaces (e.g. between different layers within a tissue) that we want to detect. To this aim we record the interferogram, i.e. the power measured by the detector as a function of the position x of mirror **B**. A filter **F** centered at 1300 nm with full-width half-maximum of 1 nm can be placed on the optical path. **D** is a detector (in a real OCT system it would be a camera).

First assume that **A1** and **B** fully reflect and the filter **F** is not present. The power measured by the detector is proportional to the temporal average of the modulus square of the total electric field,

$$P_D \propto \left\langle |\mathcal{E}_A(t) + \mathcal{E}_B(t+\tau)|^2 \right\rangle = \left\{ \left\langle |\mathcal{E}_A(t)|^2 \right\rangle + \left\langle |\mathcal{E}_B(t)|^2 \right\rangle + 2\mathrm{Re}\left[\langle \mathcal{E}_A^*(t)\mathcal{E}_B(t+\tau) \rangle \right] \right\},$$

where $\mathcal{E}_A, \mathcal{E}_B$ are the fields reflected from mirrors **A**, **B** respectively, after propagation to the detector, and τ is the relatvie delay in the two arms, depending on the position x of mirror **B**, as $\tau = -2\frac{x}{c}$ (having defined $x = 0$ as the position for which the optical path difference is zero). If the mirrors have a reflectivity of 1 we can set $\mathcal{E}_A(t) = \mathcal{E}_B(t) = A\mathcal{E}_0(t)$, where $\mathcal{E}_0(t)$ is the field at the input of the interferometer, and we retrieve the usual expression for a balanced interferometer

$$\begin{aligned} P_D &\propto |A|^2 \left\langle |\mathcal{E}_0(t) + \mathcal{E}(t+\tau)|^2 \right\rangle \\ &= 2|A|^2 \left\langle |\mathcal{E}_0(t)|^2 \right\rangle \left\{ 1 + \mathrm{Re}\left[\frac{\mathcal{E}_0^*(t)\mathcal{E}_0(t+\tau)}{\langle |\mathcal{E}_0(t)|^2 \rangle} \right] \right\} \\ &= 2|A|^2 \left\langle |\mathcal{E}_0(t)|^2 \right\rangle \left\{ 1 + \mathrm{Re}[g^{(1)}(\tau)] \right\} \end{aligned} \tag{1}$$

where $g^{(1)}(\tau)$ is the degree of first-order temporal coherence. For a source with a Lorentzian spectrum it is given by $g^{(1)}(\tau) = e^{i\omega_0\tau}e^{-|\tau|/\tau_c}$ where $\omega_0 = 2\pi\frac{c}{\lambda_0}$ is the central frequency and $\tau_c = \frac{\lambda_0^2}{\pi c \Delta\lambda_{\mathrm{FWHM}}}$ is the coherence time. Therefore for a balanced interferometer $P_D \propto 1+\cos(\omega_0\tau)e^{-|\tau|/\tau_c}$ and the interference is nearly complete (the two fields can cancel eachother giving a zero output power for certain delays).

1. *[3 points]:* Calculate the maximum visibility of the interferogram in the case where the interface **A1** reflects 50% of the power and **A2** does not reflect. The visibility is defined as $V = \frac{P_{\max}-P_{\min}}{P_{\max}+P_{\min}}$, where $P_{\max}$ and $P_{\min}$ are the maximum and minimum power measured by the detector when varying x.

2. *[3 points]:* Make a qualitative sketch of the interferogram in the case where both **A1** and **A2** reflect 50% of the power, their distance is 250 μm and no filter is present

3. *[2 points]:* How does the interferogram in *2.* change if the filter **F** is inserted in the system at the position indicated in the figure?

4. *[1 point]:* If you need to increase the axial resolution (i.e. the minimum distance between **A1** and **A2** that can be measured), how would you change the source?

5. *[1 point]:* We now replace the LED with a quantum source, which produces one photon at a time (single-photon source), with the same spectrum as the LED. **D** is replaced with a single-photon detector. Discuss how the answers to questions *1.*, *2.* and *3.* will change.

End of question.

3 Woodwind instruments [10 points]

Bart Zonneveld

This problem involves the class of woodwind musical instruments, in particular the transverse flute, the clarinet and the saxophone[3]. We will analyze how the different instruments will make their sound. This problem is based on theory from the book by Fletcher & Rossing "The Physics of Musical Instruments", which I can highly recommend for the music-playing physicist (after the competition of course!).

We start by analyzing the wave propagation of sound wave through the air. We assume that both density and pressure fluctuations are small (so $\mathrm{d}\rho \ll \rho_0$ and $\mathrm{d}p \ll p_0$), so we only consider the lowest order approximation. Furthermore, we assume the air to behave adiabatically, so following the adiabatic equation

$$pV^\gamma = \text{constant},$$

where γ is some constant depending on the properties of the air.[4]

1. *[2 points]*: Use the adiabatic equation above and Newton's laws to derive the pressure wave equation

   $$\frac{\partial^2 p}{\partial t^2} = c^2 \nabla^2 p,$$

   and express the speed of sound c in terms of γ, the (mean) air density ρ_0 and the (mean) air pressure p_0.

   Here ∇^2 is the Laplace differential operator, which is given by

   $$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

   in Cartesian coordinates, and by

   $$\frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2} + \frac{\cot(\theta)}{r^2}\frac{\partial}{\partial \theta} + \frac{\partial^2}{\partial \phi^2}$$

   in spherical coordinates.

   Hint: follow an infinitesimal package of air as it is displaced and consequently its volume fluctuates under pressure fluctuations.

We now verify our assumption of adiabatic behaviour. For this assumption to be valid, we need that the heat transfer to be small, specifically $\left|\frac{\partial Q}{\partial t}\right| \ll \rho_0 C_v \left|\frac{\partial T}{\partial t}\right|$.

2. *[1 point]*: Use the fact that the heat transfer $\left|\frac{\partial Q}{\partial t}\right|$ is proportional to $\nabla^2 T$ to show that the adiabatic assumption is valid below some frequency ω_0.

In practice, this cut-off frequency is very high ($\approx 10^{10}$ Hz), so the adiabatic assumption is always satisfied.

We will now focus on the three instruments: the transverse flute, the clarinet and the saxophone. All three instruments have holes along their tubes, which effectively shorten the tubes. These holes can be closed using fingers or keys, creating lower pitched sounds. The lowest note and its overtones are thus determined by the full length of the instrument.
There are 3 properties that make the instruments different:

i The flute has two open sides, which mean that there are no pressure fluctuations on either ends. The clarinet and saxophone are open at the bell at the bottom, but are closed at the top, meaning ∇p vanishes there.

[3] Contrary to what the name might suggest, woodwind instruments are not necessarily made from wood. In fact, their defining property is that they produce sound by either spilling air using a sharp edge, or using a reed.

[4] For regular air $\gamma \approx 1.4$, but please keep γ as a parameter in your solutions.

ii The flute and the clarinet have cylindrical bores, while the saxophone has a conical bore.

iii The flute uses a sharp edge to create fluctuations, while the clarinet and the saxophone use reeds.

3. *[2.5 points]*: Use the first two properties above to show that the lowest note and its overtones of these instruments are given by:

   - Flute and saxophone: $f = \frac{c\,n}{2L}$
   - Clarinet: $f = \frac{c\,(2n-1)}{4L}$

   Here L is the instrument length and $n = 1, 2, 3, \ldots$ enumerates the lowest note and its overtones.

   Hint: For the saxophone, the notes are given by a spherical symmetric solution to the wave equation $p(t, r)$. The functional form of this equation can easily be guessed, considering that the transported power of the fluctuations is proportional to the pressure fluctuations squared.

We have computed the resonance frequencies of the instruments, but the question remains how the instrument actually produces these frequencies. We will consider only the reeds, leaving the flute aside.
The reed has its own resonance frequency, but this is not at all the correct frequency for the instrument.[5] There is thus another mechanism that ensures the reed will create the correct frequencies.
The reed controls the flow of air into the instrument according to the pressure difference inside and outside the instrument: If the pressure inside is larger than outside, the reed will bend outwards, further opening the mouthpiece and thus increasing the flow. On the other hand, if the pressure inside is smaller, the reed bends inward, reducing the airflow.[6]

4. *[1 point]*: Explain qualitatively how this property leads to the reed producing the correct frequencies in the instrument.

In the ideal case, sound waves moving down the instrument perfectly reflect at the bottom of the instrument.[7]
This reflection is due to the change of acoustic impedance in the tube, which is given by $Z = \frac{\rho c}{A}$ where A is the cross-sectional area of the tube. When the acoustic impedance changes, we get reflection of a part of the wave's power. This reflection fraction is given by $(Z_2 - Z_1)^2/(Z_2 + Z_1)^2$, where Z_1 and Z_2 are the impedances before and after the change.
Indeed, a naive approach would be to set $A = \infty$ at the end of the instrument, giving $Z_{\text{end}} = 0$ and perfect reflection. In practice, Z_{end} will be significantly smaller than Z_{tube}, but non-zero. Still this means a lot of reflection. The clarinet and the saxophone both have a bell at the bottom that helps with impedance-matching to reduce this reflection somewhat.
We will very crudely approximate the bell of the instrument by a short tube of twice the area as the original tube. In this model we thus have two impedance changes.

5. *[1.5 points]*: Neglecting multiple reflections and keeping only terms linear terms in Z_{end}, show that the instrument in this approximated bell model radiates 16/9 times as much power as the instrument without bell.

The term 'impedance' might ring a bell from electronics. In fact, we can find an analogy to electrical circuits, where the pressure translates to electrical potential and the airflow translates to electrical current.

6. *[0.5 points]*: The mass of the air in a confined tube acts as an 'acoustic component' with impedance (per unit length) of $Z = \frac{\rho c}{A}$. In the electrical analogy, does this 'component' correspond to a resistor, a capacitor or an inductor? Explain your answer using physical arguments, so not just based on the complex phase of the impedance.

7. *[1 point]*: What physical systems in or attached to a woodwind instrument would correspond to the other two components? Please explain your answers.

[5] In fact, exciting this frequency gives a high pitch squeak, sometimes heard at beginners. Sorry neighbors...
[6] In some cases blocking the flow completely.
[7] Although we need this reflection to get resonance, perfect reflection would mean that the sound never reaches our ears!

By opening keyholes, the musician can create an open connection to the outside, somewhere in the instrument. In the ideal situation, this forces the pressure at that point to be zero, effectively shortening the instrument and thus creating a higher pitch. In practice however, the air inside the open keyhole, which is in between the tube and the 'outside world' also has some mass, and thus some impedance.

8. *[0.5 points]*: Using the electrical analogy, argue that a realistic open keyhole acts as a high-pass filter.

On the one hand, this effect is annoying: high overtones will not correspond to the (first) open keyhole, instead they become overtones of the lowest tone of the instrument. On the other hand, this effect is actively used at the 'octave key', which can open a small hole (so large impedance), which suppresses only the lower notes, forcing the instrument to play one octave higher.

End of question.

4 Death Race pilot [10 points]

Prof. Dr. Alexei Lyulin

A Death Race pilot traveling at speed v_0 suddenly sees a long, high wall across the road ahead. To avoid a collision, the pilot can either

1. *[1 point]:* Slam on the brakes,
2. *[2 points]:* Simply swerve (without breaking) to the side (his trajectory will form a circle in this case), or
3. *[7 points]:* Swerve to the side while simultaneously breaking with the rear wheels. In this case, using steering, the friction force which acts on the rear wheels, can change its direction. The pilot trajectory is still assumed to be circular.

Which of these three methods is more effective (and by how much?), i.e., which method would be preferable to avoid a collision with the nearest obstacle? To estimate this efficiency, calculate, in each case, the distance L (from the wall) between the start of the pilot action and the full stop of the auto. Total mass (pilot + car) is M. The coefficient of friction between the wheels of the racer and the road is μ. Assume that the total weight (pilot + car) is equally distributed between front and rear wheels.

End of question.

5 Gravitational waves of a binary system [10 points]

Prof. Dr. Gijs Nelemans

One of the consequences of Einstein's General Relativity (GR) theory is that waves in spacetime itself are possible: gravitational waves (GW). In this exercise we will investigate some of the properties of these waves, in part by comparing them to waves in a string, a concept that you should be familiar with. In GR, gravitational waves are caused by accelerated masses, similar to the production of electromagnetic waves by accelerated charges. However, conservation considerations show that only a changing mass quadrupole moment generates GW. Considering a sources consisting of two objects orbiting around a common center of mass with masses m_1, m_2, separation a and angular frequency ω, the energy lost in the emission of GW is given by

$$\dot{E}_{GW} = \frac{32}{5}\frac{G}{c^5}\mu^2 a^4 \omega^6$$

with $\mu = \frac{m_1 m_2}{m_1+m_2}$.

1. *[1 point]:* Use Kepler's third law to simplify this equation by eliminating a and use $\omega = \pi f_{GW}$ to write it in terms of the frequency of the GW (f_{GW}) that is twice the frequency of the binary. The combination of mass terms $(m_1 m_2)^{3/5}/(m_1 + m_2)^{1/5} = \mathcal{M}$ is called the chirp mass.

2. *[1 point]:* Derive the equation for the GW flux (energy per unit time per unit area) that passes through at distance d of the source.

3. *[2 points]:* However, contrary to for instance CCD camera's (such as in your phone), we don't detect GW via the energy in the waves, but we measure the distortion of spacetime, which essentially is the amplitude of the wave. We first recap the case of waves in a string. Use dimensional analysis to derive how the energy of waves in a string depend on the properties of the string (mass or density) and the wave (amplitude, frequency).

4. *[2 points]:* Discuss what is different in the case of GW and derive how for GW the energy depends on the properties of the GW (amplitude, frequency). Derive what combination of natural constants ($G^a c^b$) should be the pre-factor to make the units work out correctly.

5. *[1.5 points]:* Equate the energy flux at distance d to this energy flux in the GW to determine how the amplitude of the GW (tradionally given in strain amplitude $h = \Delta L/L = A/L$) depends on the properties of the source and the distance.

6. *[1 point]:* Calculate the value of the strain for a number of different types of binary sources (note the different units for distance!):

   - The first detected GW source, GW150914 consisting of two black holes with masses m_1, $m_2 = 36, 29\ M_\odot$ at $f_{GW} = 100$ Hz and $d = 440$ Mpc
   - The first detected double neutron stars GW170817, with masses m_1, $m_2 = 1.4, 1.3\ M_\odot$ at $f_{GW} = 100$ Hz and $d = 40$ Mpc
   - A binary consisting of two white dwarfs, with masses m_1, $m_2 = 0.6, 0.6\ M_\odot$ at $f_{GW} = 2 \cdot 10^{-3}$ Hz and $d = 10$ kpc
   - A binary consisting of two massive black holes, with masses m_1, $m_2 = 106, 106\ M_\odot$ at $f_{GW} = 2 \cdot 10^{-3}$ Hz and $d = 10$ Gpc

   For this question, you may use that 1 Pc $= 3.0857 \cdot 10^{16}$ m and $M_\odot = 1.988 \cdot 10^{30}$ kg.

7. *[1.5 points]:* The first two were measured with detectors on Earth (the LIGO and Virgo detectors) that have arm lengths of 4 and 3 km, but effective arm lengths (L) of order 500 km. Calculate the difference in armlength ΔL that the GW cause in the detector and compare that to the wavelength of the lasers used (about 1 μm) and discuss how challenging you think this is.

End of question.

6 The Penrose process [10 points]

Dr. Beatrice Bonga

Rotating black holes have a special region called the **ergoregion**, which is located between the event horizon $r_+ = M + \sqrt{M^2 - a^2}$ and the ergosurface $r_{\rm erg} = M + \sqrt{M^2 - a^2\cos^2(\theta)}$, where M is the mass of the black hole and $a = J/M$ describes its spin (the expressions are given in Boyer-Lindquist coordinates). In this region spacetime is dragged so intensely that staying stationary relative to observers far away is impossible. Within this region, all particles must rotate with the black hole, and energy can be extracted through the Penrose process. You will explore this in this exercise.

Warm-up: the BH-Powered Circuit

Figure 3 illustrates a gedankenexperiment where a ring is lowered into the ergoregion of a spinning black hole. Rigid shafts connect this ring to a magnet inside a wire coil far away.

1. *[1 point]:* Explain how in this set-up the light will turn on.
2. *[0.5 points]:* Identify the physical quantity being extracted from the black hole to power the lightbulb.

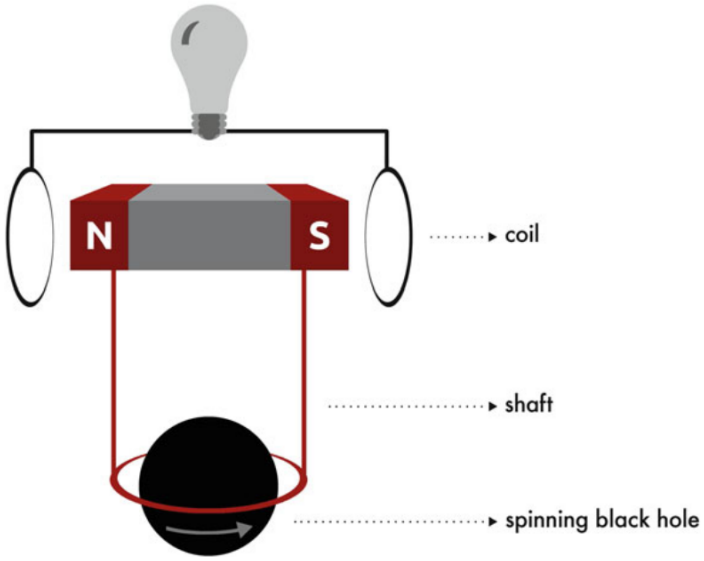


Figure 3: A gedankenexperiment in which the rotation of the black hole will turn on the light without directly touching it. Figure taken from https://link.springer.com/book/10.1007/978-3-319-19000-6.

Penrose process

Penrose came up with the following thought experiment. Consider a particle with rest mass μ_0 at rest at infinity (i.e., $E^{(0)} = \mu_0$) confined to the equatorial plane and falling into the rotating black hole. The radial evolution of such a trajectory is described by the following differential equation:

$$\dot{r}^2 = \frac{E^2}{\mu^2} + \frac{2M}{r^3}\left(\frac{aE}{\mu} - \frac{L}{\mu}\right)^2 + \frac{a^2E^2 - L^2}{\mu^2 r^2} - \frac{r^2 + a^2 - 2Mr}{r^2}$$

where the dot indicates a derivative with respect to the affine parameter along the trajectory, E is the energy of the particle, L its angular momentum and μ its rest mass. At a turning point in its trajectory, r_0, the particle decays into two products with masses $\mu_{\rm fin}$. This is illustrated in Fig. 4.

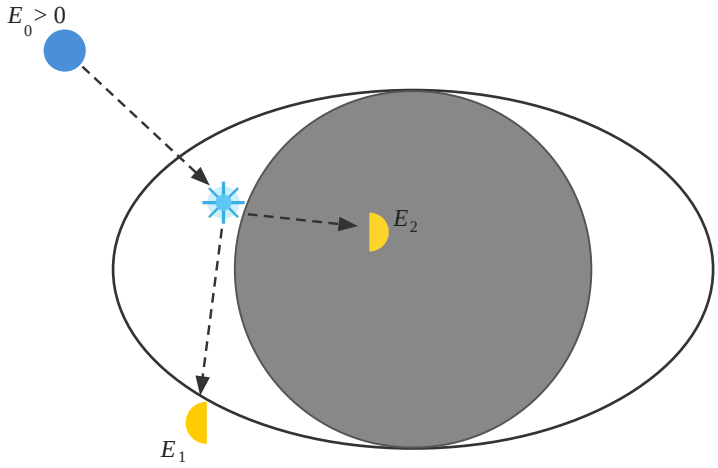


Figure 4: Illustration of the Penrose process: one particle is on a geodesic trajectory arriving from far away and at its turning point decays into two particles with $E^{(1)} > E^{(2)}$.

3. *[4 points]:* Show that the energy of the escaping products is given by:

$$E^{(1)} = \frac{\mu_0}{2}\left[1 + \sqrt{\frac{2M}{r_0}\left(1 - \frac{4\mu_{\rm fin}^2}{\mu_0^2}\right)}\right]$$
$$E^{(2)} = \frac{\mu_0}{2}\left[1 - \sqrt{\frac{2M}{r_0}\left(1 - \frac{4\mu_{\rm fin}^2}{\mu_0^2}\right)}\right].$$

   Hint: use angular momentum and energy conservation.

4. *[3 points]:* Derive the condition on r_0 such that $E^{(1)} > E^{(0)}$. If $E^{(1)} > E^{(0)}$, what is the energy of the second particle ($E^{(2)}$)?

5. *[1.5 points]:* Now let's assume the particle decays into two photons (so that $\mu_{\rm fin} = 0$). Show that the maximum efficiency $\eta = \frac{E^{(1)} - E^{(0)}}{E^{(0)}}$ for an extremal black hole (that is, a black hole that rotates at the maximum spin for which $a = M$ and $r_0 = r_+$) is

$$\eta_{\rm max} = \frac{\sqrt{2} - 1}{2}.$$

End of question.

7 Rotating elastic cord [10 points]

Prof. Dr. Alexei Lyulin

A bead of mass M is attached to the end of an elastic cord of mass m, whose length in the undeformed state is L_0. The cord with the bead rotates in a horizontal plane about an axis that passes through the other end of the cord with some angular velocity ω. The bead slides on a smooth surface; the cord does not sag. The additional stretching of the cord due to a change in its cross-sectional area S can be neglected. The cord obeys Hooke's law for any deformation and its Young's modulus is given by E.

1. *[7 points]:* How does the distance L from the axis of rotation to the bead depend on the angular velocity ω?
2. *[3 points]:* Analyse this dependence for slow rotation ($\omega \to 0$). Produce the equation from which the critical angular velocity ω_{cr} (when $L \to \infty$) can be calculated.

End of question.

8 Wave function collapse [10 points]

Prof. Dr. Gesa Welker

Researchers worldwide are trying to find the limits of Quantum Mechanics, for example by experimentally testing wave function collapse models. We consider a mechanical resonator that is used in such an experiment. It can be described by the Hamiltonian

$$H = \hbar\Omega_m \hat{b}^\dagger \hat{b} + \frac{1}{2}\hbar\Omega_m, \tag{2}$$

where Ω_m is the mechanical resonance frequency of the fundamental resonator mode and $\hat{b}^\dagger$ and $\hat{b}$ are the phonon creation and annihilation operators with

$$\hat{b} = \sqrt{\frac{m\Omega_m}{2\hbar}}\left(\hat{x} + \frac{i}{m\Omega_m}\hat{p}\right) \tag{3}$$

and $\hat{b}^\dagger$ its complex conjugate. m is the resonator mass.

1. *[2 points]:* Explain qualitatively (in 1-2 sentences) what wave function collapse is. Then name 2 potential causes of wave function collapse. For each of those causes, what are the consequences for the foundations of Physics?

2. *[3 points]:* For some wave function collapse experiments, the resonator needs to be cooled into its quantum mechancial ground state. Using the equipartition theorem, calculate the effective temperature that the resonator needs to have to be in the quantum-mechanical ground state.

3. *[3 points]:* Using equation 3, calculate the amplitude of the resonator's zero-point fluctuations. What are the implications of this value for measuring quantum-mechanical superpositions in position?

4. *[2 points]:* In some collapse models (e.g. the so-called CSL model), wave function collapse will create heat, which shows up as extra noise in the displacement of the resonator. This noise can fluctuate and be different depending on its frequency. Experiments therefore measure the mechanical noise spectrum of the resonator, see Figure 5 as an example. The displacement noise spectral density is defined as the Fourier transform of the displacement autocorrelation

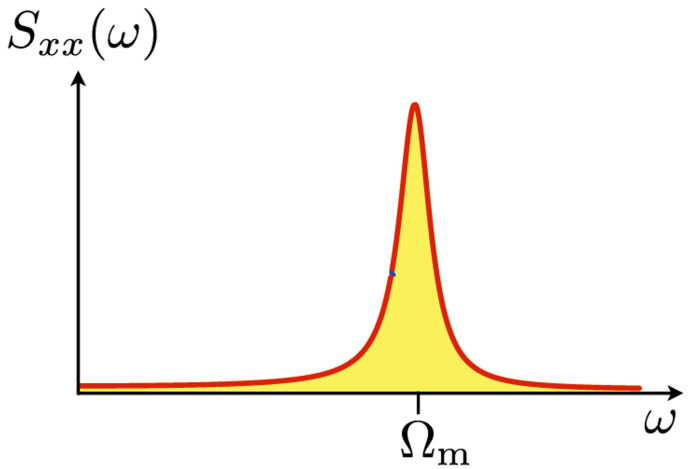


Figure 5: Mechanical noise spectrum of the resonator, graph adapted from arXiv:1303.0733v1

$$S_{xx}(\omega) \equiv \int_{\infty}^{-\infty} \langle x(t)x(0)\rangle e^{i\omega t} \mathrm{d}t \tag{4}$$

How can you derive the variance of the mechanical displacement, $\langle x^2 \rangle$, from equation 4 and a measured mechanical noise spectrum?

End of question.

9 Optical waveguiding principles [5 points]

Prof. Dr. Oded Raz

A planar optical waveguide has a **core** (film) refractive index n_f. On one side (or symmetrically both sides, if you prefer) the core is surrounded by N cladding layers. The refractive index of cladding layer k is n_k and its thickness is d_k for $k = 1, 2, \ldots, N$. Beyond the last layer is a semi-infinite external medium (the "outer cladding") with refractive index n_{N+1}. The launch medium has reflective index n_0 (often $n_0 = 1$ for air).

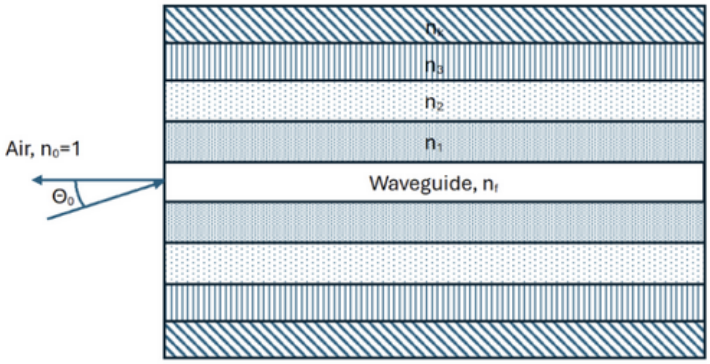


Figure 6: An example of the setup.

Assume $n_f > n_k$ for all cladding layers so that guidance by total internal reflection is possible. In the simplest single-cladding case, the critical angle condition is $\theta_c = \sin^{-1}(n_1/n_f)$.

The beam enters the waveguide through the input facet at an angle θ_0 (in medium n_0) measured relative to the facet normal.

1. *[1.5 points]:* As a warmup, assume a single cladding layer. Using Snell's law and total internal reflection, show that the **maximum acceptance half-angle** satisties

$$\theta_{\max} = \sin^{-1}\left(\frac{\sqrt{n_f^2 - n_1^2}}{n_0}\right)$$

   and define the numerical aperture

$$\mathrm{NA} = n_0 \sin(\theta_{\max}) = \sqrt{n_f^2 - n_1^2}.$$

   (These are the standard ray-optics acceptance-angle/NA relations).

2. *[3.5 points]:* Now assume N cladding layers that are optically thick enough (or the field penetration is negligible) so that no frustrated-TIR tunneling occurs through the multilayer stack. In that case, the multilayer cladding behaves (for confinement purposes) like an effective semi-infinite cladding whose index is the largest index encountered outside the core:

$$n_{\text{clad, max}} = \max\{n_1, n_2, \ldots, n_{N+1}\}.$$

   Show that a closed-form estimate for the acceptance angle is

$$\theta_{\max} \approx \sin^{-1}\left(\frac{\sqrt{n_f^2 - n_{\text{clad, max}}^2}}{n_0}\right)$$

   and therefore

$$\mathrm{NA} \approx \sqrt{n_f^2 - n_{\text{clad, max}}^2}.$$

End of question.